

lines. Each sensor interrupt increments the object count and updates the display in real time.

After completing the code, the correct Arduino board (Arduino Nano) is selected in the IDE, and the source file (Object_Counter.

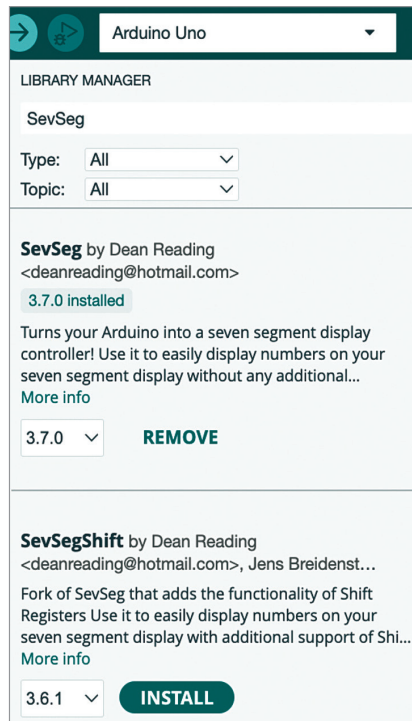


Fig. 4: Installing the library

```
#include <SevSeg.h>

#define INTERRUPT_PIN 2
#define RESET_PIN 10

#define NUM_DIGITS 3
const byte digitPins[NUM_DIGITS] = {A2, A1, A0};
const byte segmentPins[] = {3, 4, 5, 6, 7, 8, 9};

volatile unsigned long count = 0;
unsigned long lastInterruptTime = 0;
#define DEBOUNCE_DELAY 50

SevSeg sevSeg;

void setup() {
  pinMode(INTERRUPT_PIN, INPUT_PULLUP);
  pinMode(RESET_PIN, INPUT_PULLUP);

  attachInterrupt(digitalPinToInterrupt(INTERRUPT_PIN), incrementCount, FALLING);
  attachInterrupt(digitalPinToInterrupt(RESET_PIN), resetCount, FALLING);

  sevSeg.begin(COMMON_CATHODE, NUM_DIGITS, digitPins, segmentPins);
  sevSeg.setBrightness(100);
}
```

Fig. 5: Code snippet configuring the pins

ino) is uploaded. Fig. 5 shows the snippet of source code.

Construction and testing

Assemble the circuit on a general-purpose PCB or breadboard. Solder the resistors and capacitors first, followed by the voltage regulator and ULN2803A ICs. IC sockets are recommended for easy replacement. Mount the Arduino Nano using pin headers, and ensure the 7-segment displays are oriented correctly based on their pinout. Finally, solder the PNP transistors and connectors for power and sensor interfacing. The PCB may then be secured inside an enclosure using the mounting holes.

Before powering on, check component polarity carefully. Apply a 12V DC supply to the input connector only after verification. The regulator should provide a stable 5V output, and the Arduino power LED should illuminate.

Mount the sensor near the edge of the conveyor belt at the loading end, maintaining sufficient spacing between the sensor and the cement sacks for reliable detection. During testing, allow sacks to pass along

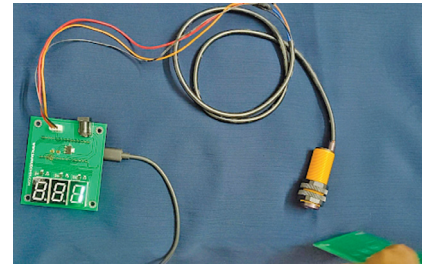


Fig. 6: The author's prototype

TO DOWNLOAD SOURCE CODE AND RESOURCES



Scan this QR code or go to <https://sourcecode.electronicsforu.com> to access and download source code and other resources for this project

the belt. Each object is detected and counted in real time, and the total is displayed. The reset pushbutton can be used to begin a new counting cycle. Fig. 6 shows author's prototype.

Installation

Install the proximity sensor near the loading end of the conveyor, close to the belt edge, so each cement sack is detected as it is placed on the belt. A side-mounted position facing the broad surface of the sack provides stable sensing and reduces dust interference. Mount the sensor at mid-sack height, typically 150-250mm above the belt, and align it perpendicular to the sack's movement, or slightly angled downward, to minimise reflections. Maintain a sensing distance of approximately 50-100mm, with at least 30mm clearance to prevent contact and false triggering.

After installation, test the system with both empty and loaded belt runs, and fine-tune the alignment to ensure consistent operation. **EFY**



A. Samiuddin is a freelance author and technology consultant